

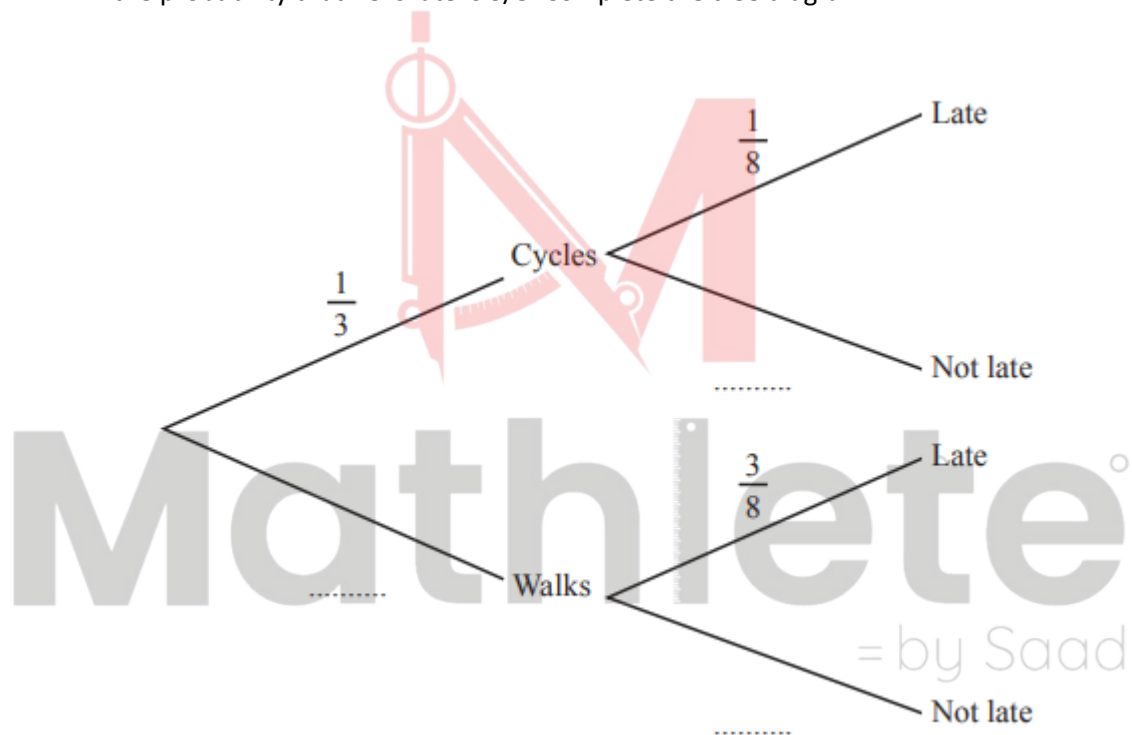
Probability

Worksheet 1 - Basic (Non-Calculator)

1. M16/qp22/q21 (0580)

Dan either walks or cycles to school. The probability that he cycles to school is $\frac{1}{3}$.

- a. Write down the probability that Dan walks to school. [1]
- b. When Dan cycles to school the probability that he is late is $\frac{1}{8}$. When Dan walks to school the probability that he is late is $\frac{3}{8}$. Complete the tree diagram.



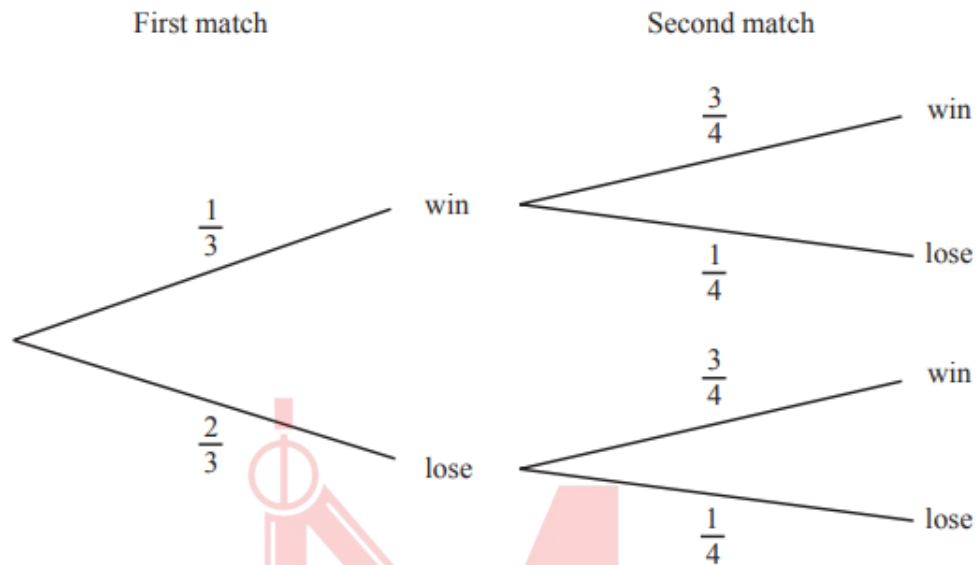
- c. Calculate the probability that [2]

i. Dan cycles to school and is late, [2]

ii. Dan is not late. [3]

2. S16/qp21/q19 (0580)

The probability of a cricket team winning or losing in their first two matches is shown in the tree diagram.



Find the probability that the cricket team wins at least one match. [3]

3. S17/qp11/q9

A bag contains red and blue pegs. There are 40 pegs in the bag. The probability of choosing a red peg from the bag is 0.4.

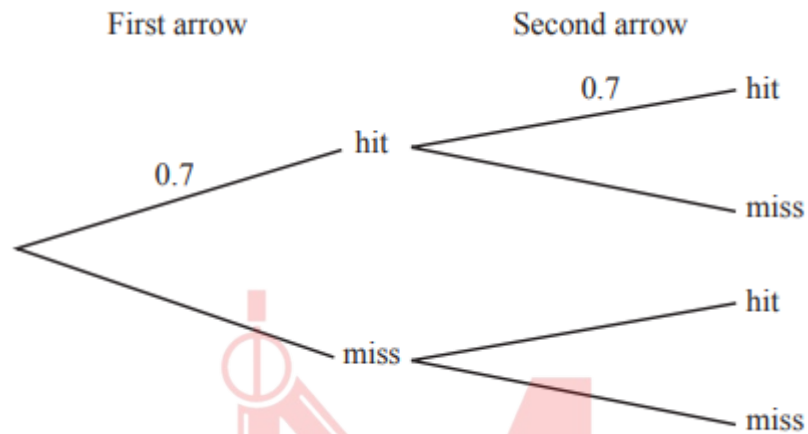
- Work out the number of red pegs in the bag. [1]
- More red pegs are added to the bag. Work out the number of red pegs that must be added to the bag so that the probability of choosing a blue peg is 0.2. [2]

4. W17/qp12/q19

Each time an archer fires an arrow, the probability that she hits the target is 0.7 . She fires two arrows.

- a. Complete the tree diagram.

[1]



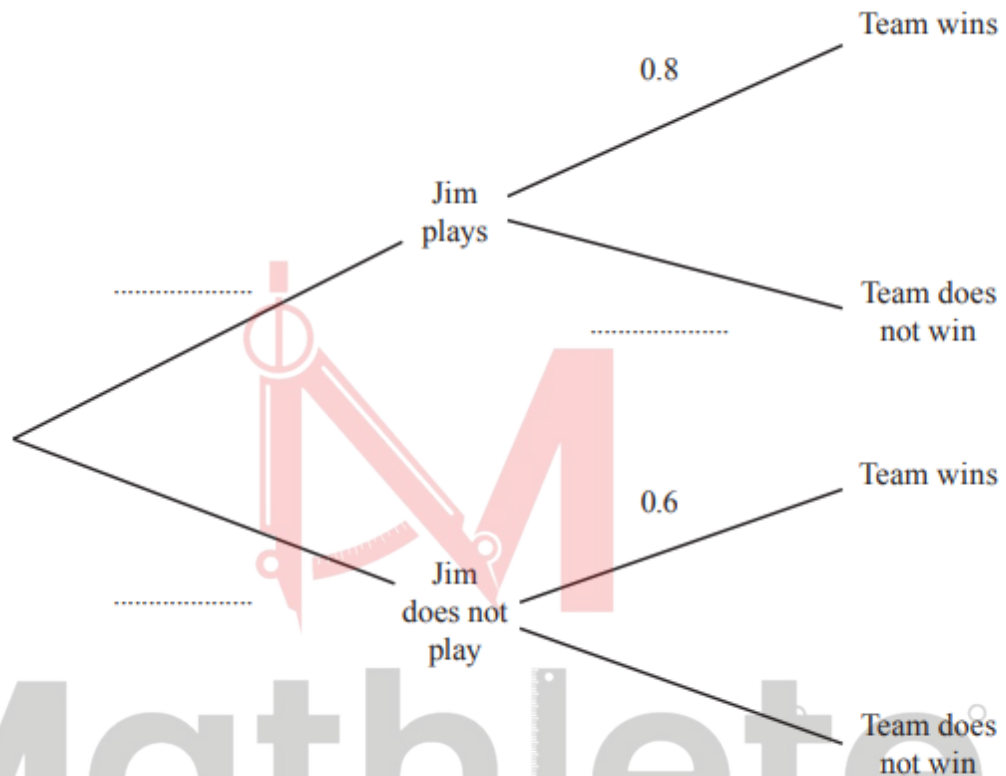
- b. Find the probability that
- i. she hits the target twice,
- ii. she hits the target exactly once.

[1]

[1]

5. S19/qp11/q18

Jim plays for his local football team. The probability that Jim plays in the next match is 0.7. If Jim plays in the match, the probability of his team winning is 0.8. If Jim does not play in the match, the probability of his team winning is 0.6.



- Complete the tree diagram. [2]
- Calculate the probability that Jim's team wins their next match. [2]

6. W19/qp21/q20 (0580)

The probability that the school bus is late is $\frac{9}{10}$. If the school bus is late, the probability that Seb travels on the bus is $\frac{15}{16}$. If the school bus is on time, the probability that Seb travels on the bus is $\frac{3}{4}$. Find the probability that Seb travels on the bus. [3]

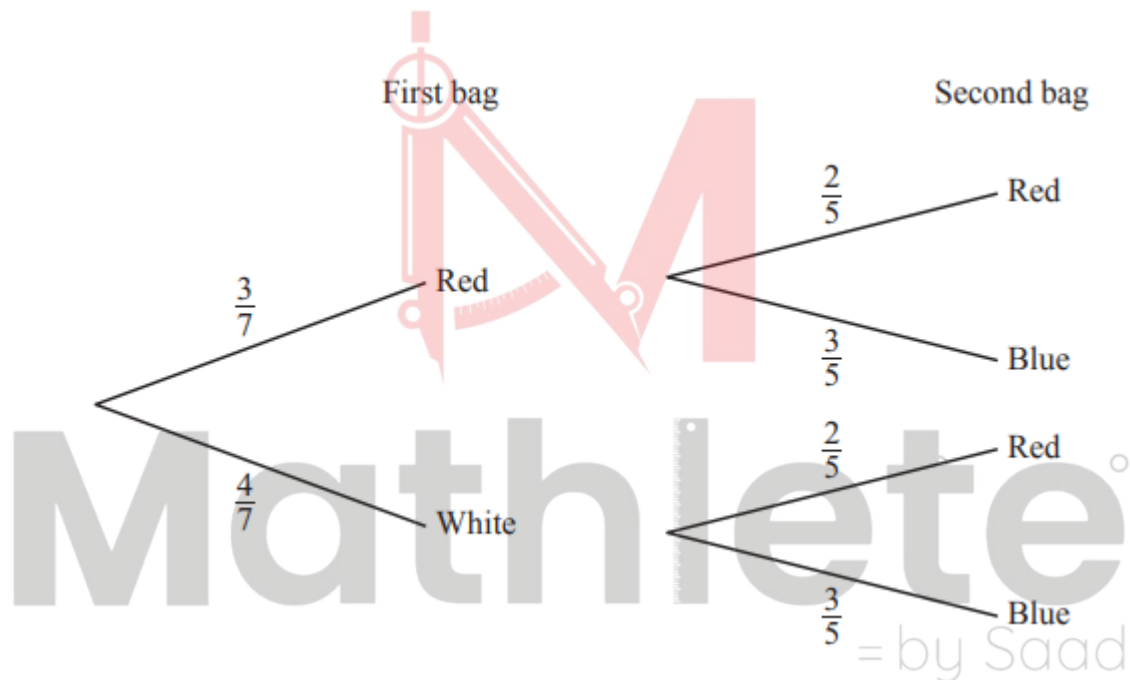
7. W19/qp22/q18 (0580)

Harris is taking a driving test. The probability that he passes the driving test at the first attempt is 0.6 . If he fails, the probability that he passes at any further attempt is 0.75 . Calculate the probability that Harris

- a. passes the driving test at the second attempt, [2]
- b. takes no more than three attempts to pass the driving test. [2]

8. W20/qp12/q15

Two bags contain beads. The first bag contains 7 beads, of which 3 are red and 4 are white. The second bag contains 5 beads, of which 2 are red and 3 are blue. One bead is taken, at random, from each bag. The tree diagram is shown below.



Find the probability that

- a. both beads are red, [1]
- b. both beads are white, [1]
- c. exactly one bead is red. [2]

9. W20/qp23/q11 (0580)

A bag contains 7 red discs, 5 green discs and 2 pink discs.

- Helen takes one disc at random, records the colour and replaces it in the bag. She does this 140 times. Find how many times she expects to take a green disc. [2]
- Helen adds 9 green discs and some pink discs to the discs already in the bag. The probability of taking a green disc is now $\frac{2}{7}$. Find the number of pink discs that Helen added to the bag. [2]

10. S21/qp11/q11

- In a survey, 3 out of every 100 women were taller than 1.9m. One of these 100 women is picked at random. Calculate the probability that she is not taller than 1.9m. [1]
- A new housing estate is being planned. There are three possible plans: A, B and C. A survey was carried out to see which plan people preferred. The relative frequency table shows the results.

Plan	A	B	C
Relative frequency	0.3	0.5	0.2

52 people preferred plan C.

- Find how many people preferred plan A. [2]
- Calculate the total number of people surveyed. [1]

11. S21/qp12/q7

A bag contains coloured counters. A counter is taken from the bag at random. The table shows the probabilities of taking a counter of each colour. Complete the table. [2]

Colour	Red	Green	Blue	Yellow
Probability	0.15	0.3		0.42

12. S21/qp23/q26 (0580)

Malik goes to a shop every day to buy bread. On any day, the probability that Malik goes to the shop in the morning is 0.7 . If he goes in the morning, the probability that there is bread for Malik to buy is 0.95 . If he goes later, the probability that there is bread for Malik to buy is 0.6 . Calculate the probability that, on any day, there is bread for Malik to buy. [3]

Answer Key:

1.(a) $\frac{2}{3}$ (b) $\frac{2}{3}$, $\frac{7}{8}$, $\frac{5}{8}$ (c)(i) $\frac{1}{24}$ (ii) $\frac{17}{24}$

2. $\frac{5}{6}$

3. (a)16 (b)80

4. (a) Probabilities 0.7 and 0.3 on the correct branches 1(b)(i) 0.49 oe (b)(ii) 0.42 oe

5.(a)0.7,0.3,0.2,0.4 (b)0.74

6. $\frac{147}{160}$

7.(a)0.3 (b)0.975

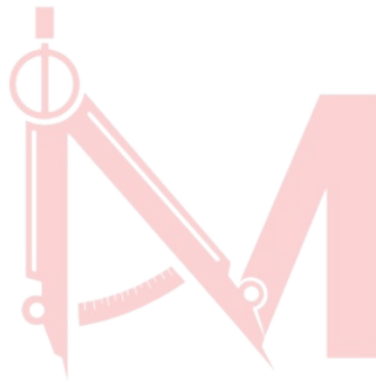
8.(a) $\frac{6}{35}$ (b)0 (c) $\frac{17}{35}$

9.(a)50 (b)26

10.(a)0.97 (b)(i)78 (ii)260

11. 0.13

12. 0.845



Mathlete
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